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Why Existence Splits: Prawitz's Ecumenical Logic and the Double-Negation Subtopos

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In Prawitz's ecumenical system [6, 7], a classical and an intuitionistic logician share conjunction, the universal quantifier, negation and absurdity, but each keeps their own disjunction, implication and existential quantifier ($\vee_c, \to_c, \exists_c$ beside $\vee_i, \to_i, \exists_i$). Derivability in the system reduces to intuitionistic derivability, yet Pereira, Pimentel and de Paiva argue that the ecumenical perspective cannot be reduced to the Gödel–Gentzen translation [6]. Why is the logical vocabulary divided exactly in this way, and what kind of unity holds the two logics together? The talk proposes a categorical answer.

Interpret formulas in the subobject lattices of an arbitrary topos $\mathcal{E}$, and let $c = \neg\neg$ be the universal closure operation of the double-negation topology. The c-closed subobjects of X form a Boolean algebra $\mathrm{Cl}(X)$, isomorphic to the subobject lattice of the associated sheaf of X in $\mathrm{Sh}_{\neg\neg}(\mathcal{E})$ [5], and the inclusion $\mathrm{Cl}(X) \hookrightarrow \mathrm{Sub}(X)$ is right adjoint to c. The shared operators all preserve closed subobjects: substitution f^*, $\wedge$, $\forall_f$, $\bot$, and $\neg$ (indeed $\neg A$ is closed for every A). Joins and $\exists_f$, which are left adjoints ($\exists_f \dashv f^*$ in Lawvere's analysis of the quantifiers [4]), do not, and must be recomputed by closure. Up to intuitionistic equivalence, the classical operators are exactly these recomputations:

$$A \vee_c B = \neg\neg(A \vee B), \qquad \exists_c x\, A = \neg\neg\, \exists x\, A, \qquad A \to_c B = A \to \neg\neg B.$$

In short, *existence splits because it is a left adjoint*; implication, itself a right adjoint, splits only through its consequent. For the single-atom formulation of [6], this semantics is sound and complete.

The reflection is oriented. $\mathrm{Sh}_{\neg\neg}(\mathcal{E})$ is a dense Boolean subtopos of every topos, whereas every subtopos of a Boolean topos is Boolean [2]: the intuitionistic layer can never appear as a reflection of the classical one inside a single universe, and hosting it requires an added modality (the S4 translation recalled in [7]). On this reading, the ecumenical system is not a translation between two languages but a reflection internal to one topos, and the unit $A \leq \neg\neg A$ measures the gap between the two existences. Since the argument uses only density, any dense Lawvere–Tierney topology j yields the same pattern, with $\exists_j = j\exists$ and $A \to_j B = A \to jB$, which suggests a family of "j-ecumenical" calculi related to Aczel's lax modalities [1].

The facts involved are standard topos theory; the contribution is to read Prawitz's division of the logical vocabulary as the trace of an adjunction. The talk ends with a short historical remark: this reading gives a precise content to Albert Lautman's notion of a *mixte* (1938), a domain in which the structure of one theory is reflected within another [3].

Keywords: Categorical logic; double-negation topology; ecumenical logic; intuitionistic logic; Albert Lautman
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